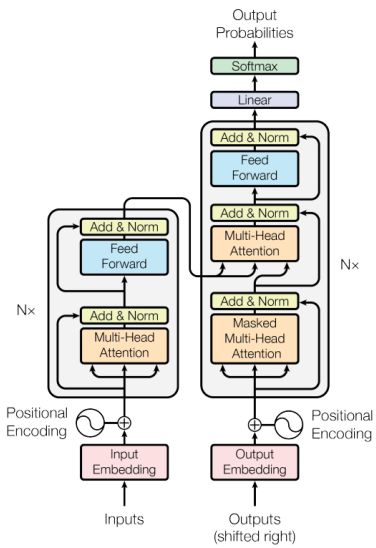
Deep Learning for NLP

Transformer
A Universal Deep Learning Architecture



Learning goals

- Understand the initial use of the Transformer
- Grasp the other application fields since then

MACHINE TRANSLATION I

- Sequence-to-sequence task
- Already served as a motivation for introducing the "ordinary" Attention-mechanism ▸ Bahdanau et al., 2015
- Crucial, that the decoder has access to the whole input sequence
  $\rightarrow$ This is very well solved by cross-attention
- Good contextualization in the encoder improves translation quality
  - (Bidirectional) RNNs/LSTMs are only (concatenated) unidirectional architectures
  - Transformer-Encoder layers are bidirectional by construction
  - Stacking them on top of each other makes this bidirectional contextualization even "deeper"

WMT 2014 EN-TO-DE AND EN-TO-FR I

Parallel training data:

Parallel data:

File	Size	CS-EN	DE-EN	HI-EN	FR-EN	RU-EN	Notes
Europarl v7	628MB	✓	✓		✓		same as previous year, corpus home page
Common Crawl corpus	876MB	✓	✓		✓	✓	same as previous year
UN corpus	2.3GB				✓		same as previous year, corpus home page
News Commentary	77MB	✓	✓		✓	✓	updated, data with document boundaries
10^9French-English corpus	2.3 GB				✓		same as previous year [md5|sha1]
CzEng 1.0	115MB	✓					same as previous year, corpus home page (avoid sections 98 and 99)
Yandex 1M corpus	121MB					✓	corpus home page; v1.3 now in original case
Wiki Headlines	7.8MB			✓		✓	Provided by CMU. The ru-en is unchanged from last year.
HindEnCorp	25MB			✓			Collected by Charles University
~~The JHU Corpus~~				~~✓~~			This is fully contained in HindEnCorp, so not made available here.

THE BLEU SCORE I

- Based on n-gram overlap from candidate and reference sentence
- Precision (for each n-gram) calculated as follows:

$$P_n = \frac{\sum_{C \in \{\text{ Candidates }\}} \sum_{n-gram \in C} \text{ Count }_{\text{clip}} (n- \text{ gram })}{\sum_{C' \in \{\text{ Candidates }\}} \sum_{n-gram' \in C'} \text{ Count } (n- \text{ gram }')}$$

- Finally, the BLEU score can be computed as

$$\texttt{BLEU} = BP \cdot \exp\left(\sum_{n=1}^{N} w_n \cdot \log(P_n)\right),$$

- where BP is a "brevity penalty" to penalize short generations, N is the number of n-grams & w_n the weight for each P_n (usually $\frac{1}{N}$)

TRANSFORMER FOR MT I

The Transformer ..

- .. outperforms the previous SOTA models
- .. at a lower number of required training FLOPs

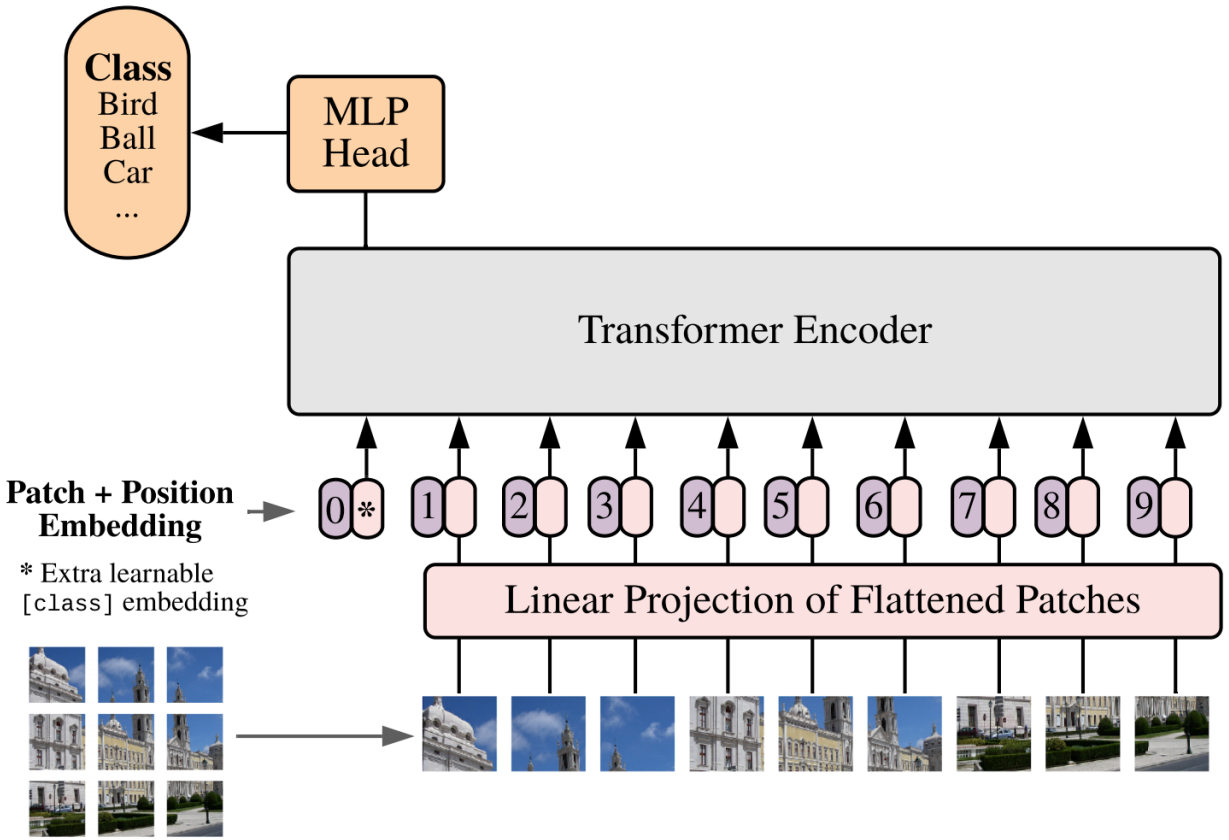
Table 2: The Transformer achieves better BLEU scores than previous state-of-the-art models on the English-to-German and English-to-French newstest2014 tests at a fraction of the training cost.

Model	BLEU		Training Cost (FLOPs)	
	EN-DE	EN-FR	EN-DE	EN-FR
ByteNet [18]	23.75			
Deep-Att + PosUnk [39]		39.2		$1.0 \cdot 10^{20}$
GNMT + RL [38]	24.6	39.92	$2.3 \cdot 10^{19}$	$1.4 \cdot 10^{20}$
ConvS2S [9]	25.16	40.46	$9.6 \cdot 10^{18}$	$1.5 \cdot 10^{20}$
MoE [32]	26.03	40.56	$2.0 \cdot 10^{19}$	$1.2 \cdot 10^{20}$
Deep-Att + PosUnk Ensemble [39]		40.4		$8.0 \cdot 10^{20}$
GNMT + RL Ensemble [38]	26.30	41.16	$1.8 \cdot 10^{20}$	$1.1 \cdot 10^{21}$
ConvS2S Ensemble [9]	26.36	**41.29**	$7.7 \cdot 10^{19}$	$1.2 \cdot 10^{21}$
Transformer (base model)	27.3	38.1	**$3.3 \cdot 10^{18}$**	
Transformer (big)	**28.4**	**41.8**	$2.3 \cdot 10^{19}$	

▸ Source: Vaswani et al., 2017

TRANSFORMER FOR COMPUTER VISION I

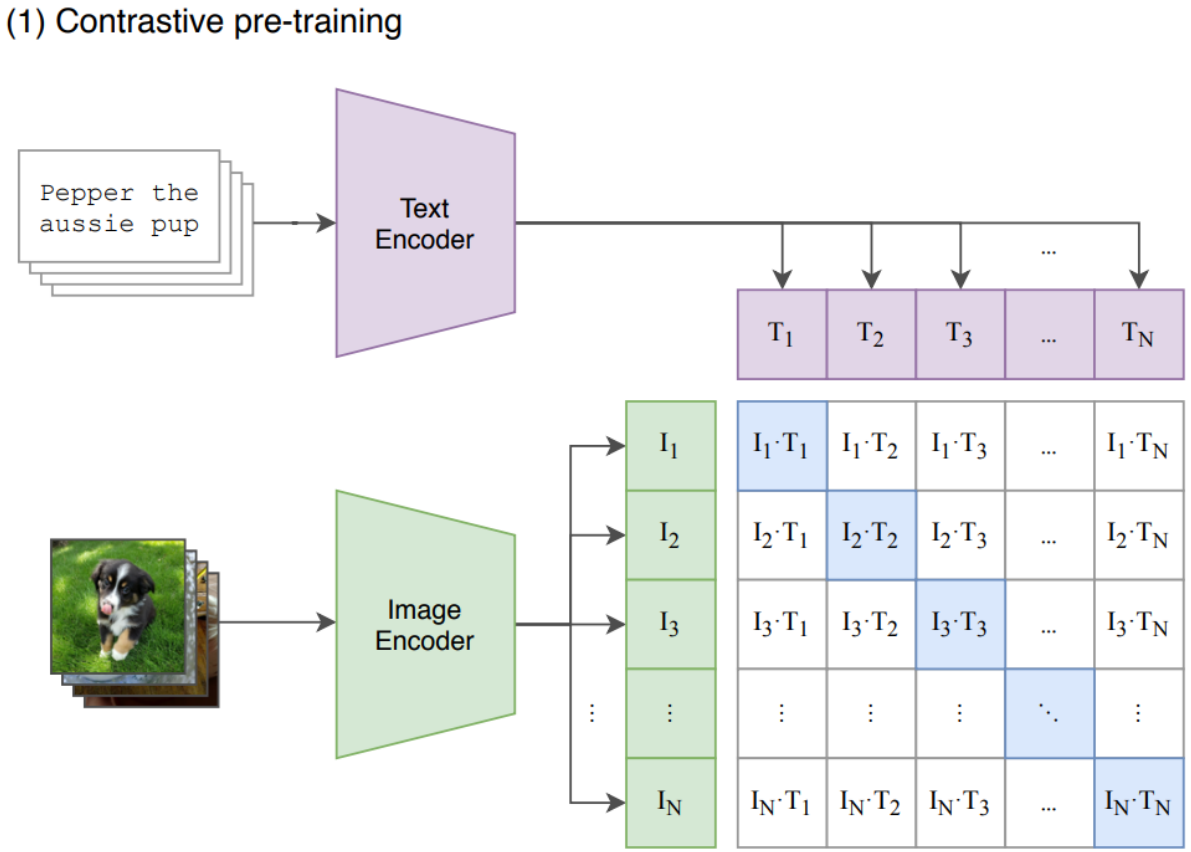
Vision Transformer



▸ Source: Dosovitskiy et al., 2020

TRANSFORMER FOR MULTIMODAL LEARNING I

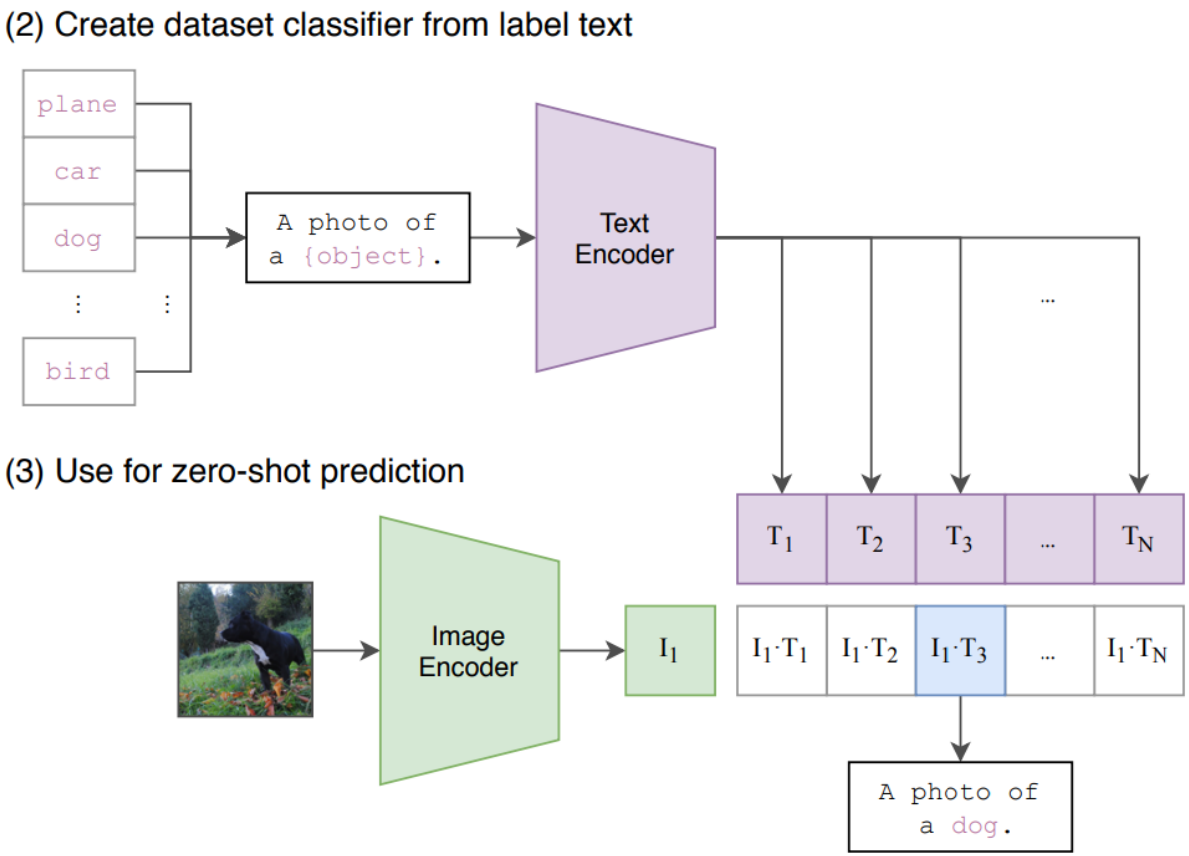
CLIP



▸ Source: Radford et al., 2021

TRANSFORMER FOR MULTIMODAL LEARNING I

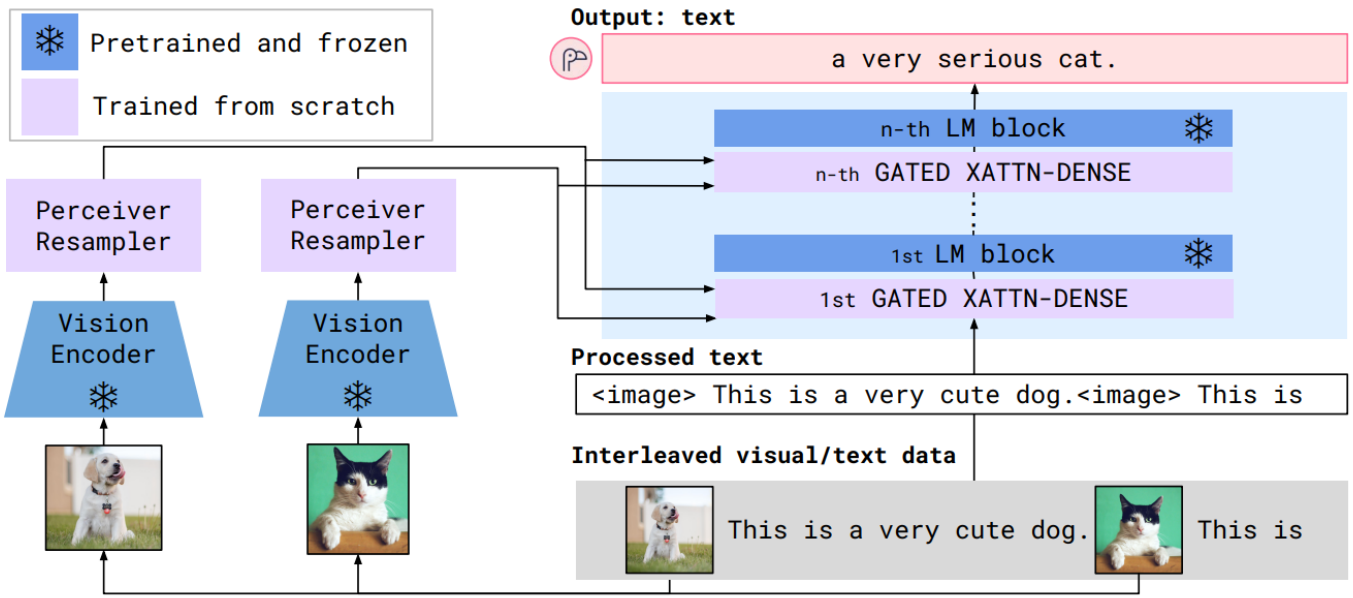
CLIP



▸ Source: Radford et al., 2021

TRANSFORMER FOR MULTIMODAL LEARNING I

FLAMINGO



▸ Source: Alayrac et al., 2022

TRANSFORMER FOR MULTIMODAL LEARNING I

GPT-4

Example of GPT-4 visual input:	
User	What is funny about this image? Describe it panel by panel.
	 Source: https://www.reddit.com/r/hmmm/comments/ubab5v/hmmm/
GPT-4	The image shows a package for a "Lightning Cable" adapter with three panels. Panel 1: A smartphone with a VGA connector (a large, blue, 15-pin connector typically used for computer monitors) plugged into its charging port. Panel 2: The package for the "Lightning Cable" adapter with a picture of a VGA connector on it. Panel 3: A close-up of the VGA connector with a small Lightning connector (used for charging iPhones and other Apple devices) at the end. The humor in this image comes from the absurdity of plugging a large, outdated VGA connector into a small, modern smartphone charging port.

Table 3. Example prompt demonstrating GPT-4's visual input capability. The prompt consists of a question about an image with multiple panels which GPT-4 is able to answer.

▸ Source: OpenAI, 2023